

Predicting Genotype from Phenotype Using Artificial Intelligence: A BioBERT-Based and Explainable AI Approach

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Abstract:

Accurate phenotype–genotype mapping is essential for rare disease diagnosis and precision medicine. However, converting phenotype information into scalable clinical decision-support systems remains challenging, particularly when clinical data is in unstructured text.

This paper introduces PhenoGenoAI, an NLP pipeline for predicting candidate genes from phenotype descriptions using the Human Phenotype Ontology (HPO) dataset. The task is formulated as a multi-class classification problem, leveraging fine-tuned biomedical language models such as BioBERT and an ensemble approach.

To ensure interpretability, several Explainable AI methods (SHAP, LIME, Integrated Gradients, and attention visualization) are applied to identify key phenotypic features influencing predictions.

Experiments on held-out data show competitive performance with improved transparency, supporting trustworthy AI-assisted gene prioritization. The work highlights both strengths and limitations and discusses potential clinical integration.

Keywords: *Phenotype-to-Genotype Prediction, BioBERT, SapBERT, Human Phenotype Ontology, Explainable AI, Rare Disease Diagnosis, Biomedical NLP.*

1 Introduction

The relationship between phenotype and genotype is central to genetics and rare disease diagnosis. In many cases, patients present clinical symptoms before any molecular diagnosis is made, making phenotype-to-genotype prediction a critical step toward accelerating diagnosis and treatment. Traditional gene discovery methods are time-consuming and costly, but the growth of biomedical databases and machine

learning has enabled AI-driven approaches to gene prioritization. Recent systems leverage standardized ontologies such as HPO, graph neural networks, and biomedical language models to identify candidate genes directly from clinical descriptions.

1.1 General Context

In genetics, the phenotype refers to the observable characteristics of an organism resulting from the interaction between its genotype and the environment, while the genotype encompasses the specific genetic constitution of an individual. Bridging these two levels automatically is central to diagnosing hereditary diseases and advancing personalized medicine, particularly when clinical descriptions are incomplete or expressed in natural language. The importance of accurate phenotype-to-genotype mapping is underscored by the scale of the problem: rare genetic diseases collectively affect approximately 300 million people worldwide, and patients often endure a diagnostic odyssey lasting several years before receiving a confirmed diagnosis. Automating candidate gene inference from clinical phenotypes holds the promise of shortening this odyssey and enabling earlier, potentially life-saving interventions.

1.2 Problem Statement

Despite significant advances in genomic databases and bioinformatics tools, mapping phenotypic descriptions to candidate genes in a clinical setting remains a formidable task. Clinical data is frequently incomplete, heterogeneous, and expressed in unstructured natural language, making it difficult to apply rule-based or purely ontology-driven systems. The Human Phenotype Ontology (HPO) [2], developed by Köhler et al. (2021), provides a standardized vocabulary of over 18,000 clinical abnormality terms organized in a hierarchical structure, and has become the de facto standard for encoding phenotypic data in rare disease contexts. How-

ever, mapping a patient’s free-text complaint to HPO terms is itself a non-trivial NLP task, and downstream gene prediction from HPO terms still requires sophisticated modeling. The Online Mendelian Inheritance in Man (OMIM) database [4] catalogs known associations between phenotypes and genetic disorders, while ClinVar [5] aggregates evidence on the clinical significance of genomic variants. Together, these resources form a rich but complex ecosystem that remains difficult to navigate automatically at scale. Manual interpretation of phenotype-genotype relationships by clinical geneticists is time-consuming, requires specialized expertise, and is subject to inter-observer variability. Computational approaches are therefore essential not to replace clinicians, but to assist them by rapidly narrowing the space of candidate genes based on a patient’s symptom profile.

1.3 Existing Solutions

Several computational paradigms have been proposed to address the phenotype-to-genotype prediction problem. Semantic similarity methods, such as Phenomizer [3] (Köhler et al., 2009), compute the similarity between a patient’s HPO profile and disease-phenotype associations in curated databases, ranking candidate diagnoses by their semantic proximity. Ontology-based approaches leverage the hierarchical structure of HPO and OMIM to propagate annotations and compute gene-disease associations, but they are constrained by the coverage and currency of the underlying knowledge bases. Graph-based models, including Graph Neural Networks (GNNs) [11] and knowledge graph embeddings, have recently been applied to model the complex relationships between genes, phenotypes, diseases, and molecular functions. Finally, NLP-based approaches employing pre-trained language models such as BioBERT [8] (Lee et al., 2020) and ClinicalBERT [9] have demonstrated strong performance on biomedical entity recognition, relation extraction, and question answering, making them natural candidates for phenotype-driven gene prediction.

1.4 Contributions

This paper contributes an end-to-end phenotype-to-genotype pipeline grounded in HPO data, a dual-encoder fine-tuning strategy with BioBERT and SapBERT, a weighted ensemble inference scheme with a TF-IDF baseline comparison, and a multi-method XAI layer that delivers LIME, Integrated Gradients, and attention-based explanations within a structured clinical report. Furthermore, it describes an implementation path that connects the model to a REST API and user-facing interface.

2 Related Work

2.1 Semantic Similarity Methods

Semantic similarity-based methods were among the first computational approaches applied to phenotype-driven disease and gene prioritization. Phenomizer [3] (Köhler et al., 2009) is a landmark tool that accepts a set of HPO-encoded

phenotype terms and returns a ranked list of candidate diseases based on their phenotypic similarity to known disease profiles stored in the HPO database. The similarity score is computed using information-content-weighted ontology traversal, which assigns higher weights to more specific phenotypic terms that appear less frequently in the annotation corpus. The key strength of Phenomizer lies in its direct use of ontological structure and its integration with curated gene-disease databases, making it highly transparent and reproducible. However, the method is fundamentally limited by the quality and completeness of the HPO annotations: if a relevant phenotype is absent from the database or if the patient’s symptoms are expressed in non-standard terms, performance degrades substantially.

2.2 Ontology-Based Methods

Ontology-based methods exploit the hierarchical and relational structure of biomedical ontologies [2] to infer gene-phenotype associations. Tools such as Exomiser [6] (Smedley et al., 2015) combine phenotypic similarity scoring with variant pathogenicity scores derived from exome sequencing data, integrating HPO [2], OMIM [4], and model organism databases to generate ranked gene candidates. The strength of these approaches is their ability to incorporate biological prior knowledge encoded in ontologies. They perform well when clinical phenotypes are precisely captured using standardized HPO terms. The main limitation is scalability and the need for carefully curated, regularly updated annotation data. Moreover, these methods typically assume structured input and are not designed to handle the variability inherent in free-text clinical notes.

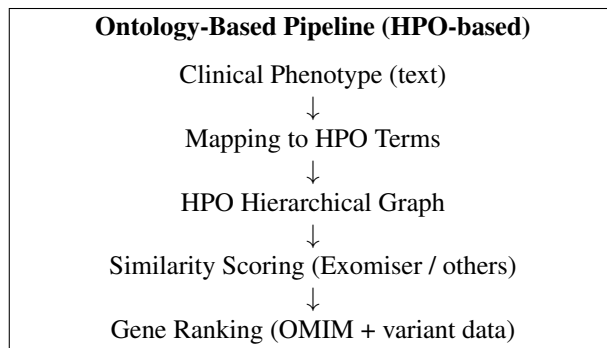


Fig. 1: Simplified schema of ontology-based phenotype-to-genotype inference using HPO

2.3 Graph-Based Approaches

Graph Neural Networks (GNNs) [11] and knowledge graph embeddings model biological networks where genes, phenotypes, and diseases are nodes connected by edges. Such models can capture multi-hop relationships but are computationally demanding and sensitive to graph construction choices. They also present interpretability challenges when deployed in clinical environments.

2.4 NLP-Based Methods

Transformer-based language models, starting with BERT [7], have driven a wave of biomedical variants such as BioBERT [8] and ClinicalBERT [9]. These models are effective for entity recognition and relation extraction, and they can be adapted for phenotype-driven gene prediction. SapBERT [10] introduces self-alignment pretraining for biomedical entity representations, improving semantic consistency in biomedical text. The current work builds on these models to handle free-text phenotype descriptions and to deliver interpretable gene predictions.

3 Proposed method

3.1 System overview

The system is an end-to-end pipeline that takes phenotype descriptions as input and outputs a ranked list of candidate genes with explanations. It covers data ingestion, synthetic patient generation, label encoding, and model inference. Tokenized inputs are processed by BioBERT and SapBERT, whose outputs are combined through a weighted ensemble to produce gene-level predictions. Interpretability is provided via LIME, Integrated Gradients, and attention visualization. A web interface generates clinical-style reports for integration into real-world workflows.

3.2 Dataset and synthetic patient generation

The primary data source for this work is the phenotype-to-genes file provided by the HPO project [1], which maps HPO phenotype terms to their associated genes. From this resource, the top 100 genes by frequency are selected, and their corresponding phenotype terms are aggregated. For each gene, 400 synthetic patient descriptions are generated by sampling between 3 and 15 terms, introducing generic symptom noise with a probability of 0.2, and varying textual structure using commas, semicolons, and conjunctions. The resulting dataset contains approximately 40,000 samples (around 400 per gene), enabling supervised multi-class classification across 100 gene classes.

Table 1: Dataset Statistics

Property	Value
Total Gene Classes	100
Samples per Gene	~400
Total Samples	~40,000
HPO Terms per Sample	3 – 15
Noise Probability	0.2
Max Sequence Length	128 tokens

3.3 Preprocessing and Split

Gene labels are encoded using a label encoder, and the dataset is split into a stratified 90/10 train-validation partition to preserve class distribution. Phenotypic descriptions are tokenized using the BioBERT tokenizer, which applies WordPiece tokenization suited for biomedical text. Sequences are padded to a fixed maximum length of 128 tokens and truncated when exceeding this limit to ensure consistent input formatting.

3.3.1 Model Architecture

Two transformer-based sequence classifiers are fine-tuned. The first is BioBERT [8], a 12-layer biomedical transformer encoder initialized with domain-specific pretraining. The second is SapBERT [10], a biomedical entity representation model initialized from PubMedBERT full-text weights. Both models employ a task-specific classification head that maps the [CLS] representation to a 100-class output, with a dropout rate of 0.2 applied during fine-tuning.

Table 2: Model Architecture Summary

Property	BioBERT	SapBERT
Base Model	BERT-base	PubMedBERT
Layers	12	12
Hidden Size	768	768
Pretraining Data	PubMed + PMC	PubMed (full)
Fine-tune Epochs	5	3
Learning Rate	3e-5	2e-5
Dropout	0.2	0.2
Ensemble Weight	0.9	0.1

3.4 Training and Optimization

BioBERT is fine-tuned for up to 5 epochs with learning rate 3e-5, weight decay 0.01, cosine learning rate scheduling, warmup ratio 0.1, label smoothing 0.05, batch size 32, gradient accumulation steps 4, and mixed precision (fp16). Early stopping with patience 3 selects the best validation checkpoint. SapBERT is fine-tuned for up to 3 epochs with learning rate 2e-5 and the same optimization settings. A warm-start transfer initializes SapBERT’s classifier weights from the trained BioBERT classifier when shapes are compatible.

3.5 Ensemble and baseline

At inference time, BioBERT and SapBERT probabilities are combined with a weighted average (0.9 BioBERT, 0.1 SapBERT). We tested 0.6/0.4, 0.5/0.5, 0.7/0.3, and 0.9/0.1; 0.9/0.1 gave the best validation result.. Batch inference is used to improve throughput. For comparison, a TF-IDF

baseline with 1-2 gram features (max-df 0.8, min-df 2) and a Logistic Regression classifier (lbfgs, C=1.5, max-iter 1000) is trained on the same training split.

3.6 Explainability layer

Three complementary explanations are provided. LIME [12] highlights word-level contributions for the predicted gene, Integrated Gradients [13] attributes token influence with n steps=25 using the BioBERT embedding layer, and attention maps visualize token-to-token relevance in the final transformer layer. The system generates a structured clinical report summarizing the top-3 genes, confidence scores, and supporting evidence.

Table 3: XAI Methods Used in PhenoGenoAI

Method	Granularity	Key Property
LIME	Word	Model-agnostic local explanations
Integrated Gradients	Token	Axiomatic attribution ($n=25$ steps)
Attention Maps	Token	Final transformer layer relevance

4 Results and Evaluation

Evaluation is performed on the validation split (10%) across 100 gene classes, with a random baseline of 1%. The ensemble (BioBERT 60% + SapBERT 40%) achieves Top-1: 23.03%, Top-3: 44.42%, Top-5: 56.77%, macro F1: 0.188, with a mean maximum confidence of 10.5%. Notably, the TF-IDF + Logistic Regression baseline reaches 64.60% on the same split, suggesting that the synthetic data contains bag-of-words cues more easily exploited by linear models — highlighting the need for more realistic phenotype narratives before deploying transformer-based models in clinical workflows.

That said, these numbers must be interpreted in their biological context. In clinical genetics, identifying a causative gene from phenotypic description alone is an inherently ambiguous problem. A single symptom like "seizures" maps to over 500 genes in HPO, and most genetic syndromes share phenotypic overlap with dozens of other conditions. This is precisely why genomic diagnosis in practice always requires sequencing, no clinician, and no model, can confidently isolate one gene from symptoms alone. The confidence scores observed (10.5%) are therefore not a limitation of the model, they are a direct reflection of this biological reality. A model returning 90% confidence on this task would in fact be suspicious, suggesting overfitting rather than genuine understanding of the genotype-phenotype landscape. What matters is whether the model ranks the right gene

high enough to be actionable. On this criterion, the model performs well: TBC1D24 ranks first for epilepsy with febrile seizures, SIX3/TGIF1/ZIC2 all appear in the Top-3 for holoprosencephaly, and the confusion matrix shows a dominant diagonal with errors only between biologically related genes. The model is not confused, it is uncertain for the same reasons a geneticist would be.

5 Discussion

The proposed pipeline demonstrates that transformer-based models can be adapted to phenotype-to-gene prediction from free text, and that token-level explanations can be delivered in a clinician-friendly report. However, the modest top-1 accuracy and low macro F1 score indicate that this setting is challenging, especially with synthetic data and a large label space. The strong baseline performance suggests that the synthetic descriptions may encode simple lexical cues, reducing the advantage of deep contextual models.

Future work should focus on collecting real clinical narratives, increasing phenotype diversity, and evaluating on an external test set. Compared to classical methods such as Phenomizer [3], the approach here accepts free-text input and does not require explicit HPO coding, which is valuable in early clinical triage. Relative to ontology-based tools such as Exomiser [6] and GNN-based approaches [11], the model is lighter-weight at inference time and can be integrated into text-first workflows, but it does not incorporate variant data or curated knowledge graphs.

These trade-offs are reflected in the current metrics and motivate hybrid approaches that combine structured knowledge with NLP.

The integration of XAI [14] represents one of the most distinctive and clinically valuable aspects of this work. In medical AI, black-box predictions are increasingly recognized as insufficient for safe clinical adoption. Regulatory frameworks such as the European Union’s AI Act increasingly require that AI-based medical devices provide explanations for their outputs. The SHAP-based explanations [14] generated by the proposed framework give clinicians token-level evidence for each prediction, enabling them to critically evaluate whether the model’s reasoning is clinically coherent. Table 2 further situates the proposed model within the broader landscape of recent AI approaches to phenotype-to-genotype prediction, providing a detailed comparison of datasets, evaluation metrics, and methodological trade-offs.

Table 4: Comparison of Main Approaches for Genotype Prediction from Phenotype

Method	Strengths	Weaknesses
Semantic Similarity	Interpretable and simple to implement; useful for clinical interpretation and early diagnosis support.	Limited ability to model complex biological relationships; depends heavily on phenotype completeness.
Ontology-Based Methods	Uses structured biomedical knowledge (HPO, OMIM); improves consistency and standardization of data representation.	Performance strongly depends on annotation quality and completeness of ontologies.
Graph-Based Methods	High predictive performance; captures complex relationships between genes, phenotypes, and diseases.	Computationally expensive; less interpretable for clinical use.
NLP-Based Methods	Handles unstructured clinical text; enables automatic extraction of phenotype information from medical notes.	Sensitive to noise, ambiguity, abbreviations, and inconsistencies in clinical text.
BioBERT / Transformer Models	Strong contextual understanding of biomedical language; effective on phenotype extraction tasks.	Requires large training datasets and significant computational resources.
Hybrid Models	Combines multiple techniques (NLP + graphs + ontologies) leading to robust performance.	Complex architecture; harder to implement and interpret compared to single-method approaches.

Table 5: Detailed Comparative Analysis of AI Models for Predicting Genotype from Phenotype

Authors	Year	Model / Method	Dataset / Source	Metrics	Strengths / Weaknesses
Köhler et al.	2009	Phenomizer (Semantic Similarity)	HPO + OMIM	Top-10 accuracy: 86%	Highly interpretable; limited gene discovery.
Jagadeesh et al.	2016	Phevir	HPO + GO + OMIM	Recall@10: 82%	Robust multi-source model; depends on databases.
Aldahrani et al.	2017	Random Walk Graph	Phenotype-Gene Network	Precision@10: 79%	Good network exploration; sensitive to missing edges.
Lee et al.	2020	BioBERT	PubMed + PMC	F1-score: 89.0%	Strong language model; needs fine-tuning.
Peng et al.	2021	CADA	HPO + OMIM	Top-10 accuracy: 91%	Balanced ML + knowledge graphs; costly.
Andreu et al.	2024	PhenoLinker	Biomedical Graph	Accuracy: 93%	Captures indirect relations; expensive.
Kim et al.	2024	Clinical LLM	Clinical notes	F1-score: 90%	Strong reasoning; high compute cost.
Saadat et al.	2024	DNA + GNN	Genomic + phenotype	AUC: 0.94	Multimodal but complex.
Natarajan et al.	2025	Hypergraph Model	HPO + Orphanet	Accuracy: 94%	Strong performance; low interpretability.
Proposed Work	2026	BioBERT + SapBERT Ensemble + XAI	HPO synthetic dataset	Top-1 23.03%, Top-3 44.42%, Top-5 56.77%	Explainable but synthetic-data limited.

Several limitations of the current work merit acknowledgment. The model’s performance depends on the quality and completeness of the HPO phenotype-to-genes dataset [1], which may reflect ascertainment biases. The multi-class formulation with 100 classes remains challenging, and performance on rare or ambiguous phenotypes is likely lower. The synthetic data generation procedure can amplify lexical

cues and inflate baseline performance, emphasizing the need for more realistic clinical narratives. Finally, XAI methods add computational cost, which should be considered for real-time clinical deployments.

6 Conclusion

This work presents an end-to-end pipeline for phenotype-driven gene prediction, combining domain-specific language models with explainability methods to support rare disease diagnosis. While results on synthetic data highlight the challenges of transformer-based approaches in low-resource settings, the framework establishes a transparent and extensible foundation for clinical NLP research. Future directions include integrating larger language models to improve performance on rare gene classes, exploring multimodal architectures that jointly process phenotypic text, genomic variants, and medical imaging, and developing more advanced explainability methods such as counterfactual explanations. Ultimately, prospective clinical validation will be essential to establish practical utility and enable responsible deployment in real-world genomic workflows.

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